

Chapter One

Part - I

Introduction to Information Security

Information Security
(SEng7432)

Introduction

- ✎ While computer systems today have some of the best security systems ever, they are more vulnerable than ever before.
- ✎ Computer and network security comes in many forms, including encryption algorithms, access to facilities, digital signatures, and using fingerprints and face scans as passwords.
- ✎ The OSI **security architecture** provides a systematic frame work for defining security attacks, mechanisms and services.
- ✎ The OSI security architecture focuses on security attacks, mechanisms and services.
 - ☞ Security attack:- Any action that compromises the security of information owned by an organization.
 - ☞ Security mechanism:- A process (or a device incorporating such a process) that is designed to detect, prevent, or recover from a security attack.
 - ☞ Security service:- A processing or communication service that enhances the security of the data processing systems and the information transfers of an organization. The services are intended to counter security attacks, and they make use of one or more security mechanisms to provide the service.

Basic Security Terminology

- ✉ **Security**, from network point of view, is the protection of the key infrastructure of the network components like computers, links between them and other devices. In general term, it is set of activities involved in protecting a valuable item like information.
- ✉ **Computer Security:-** generic name for the collection of tools designed to protect data and to hackers from attacking the organizational assets.
- ✉ **Network Security:-** measures to protect data during their transmission over the network.
- ✉ **Internet Security:-** measures to protect data during their transmission over a collection of interconnected networks(network of networks)
- ✉ **Information Security:-**means protecting **information** and **information systems** from unauthorized access, use, disclosure, disruption, modification, perusal, inspection, recording or destruction.

Why Is Computer Network and Information Security Important?

1. *To protect company assets:-* One of the primary goals of computer and network security is the protection of company assets (hardware, software and/or information).
2. *To gain a competitive advantage:-* Developing and maintaining effective security measures can provide an organization with a competitive advantage over its competition
3. *To comply with regulatory requirements and fiduciary responsibilities:-* organizations that rely on computers for their continuing operation must develop policies and procedures that address organizational security requirements.
4. *To keep your job:-* Security should be part of every network or systems administrator's job. Failure to perform adequately can result in termination.
5. *The growing fear of cyber terrorism*
6. *Attacks to the infrastructure would affect a large portion of the Internet and create a large amount of service disruption.*

Challenges in Computer Network and Information security

1. The requirements seem to be straightforward; indeed, most of the major requirements for security services can be given self-explanatory, one-word labels: confidentiality, authentication, nonrepudiation, or integrity. But the mechanisms used to meet those requirements can be quite complex, and understanding them may involve rather subtle reasoning.
2. In developing a particular security mechanism or algorithm, one must always consider potential attacks on those security features. In many cases, successful attacks are designed by looking at the problem in a completely different way, therefore exploiting an unexpected weakness in the mechanism.
3. The procedures used to provide particular security services are often counter intuitive. Typically, a security mechanism is complex, and it is not obvious from the statement of a particular requirement that such elaborate measures are needed.

Challenges in Computer Network and Information security (Cont....)

4. Having designed various security mechanisms, it is necessary to decide where to use them. This is true both in terms of physical placement and in a logical sense.
5. Security mechanisms typically involve more than a particular algorithm or protocol. They also require that participants be in possession of some secret information (e.g., an encryption key), which raises questions about the creation, distribution, and protection of that secret information.
6. It is a battle of wits between a perpetrator who tries to find holes and the designer or administrator who tries to close them. The great advantage that the attacker has is that he or she need only find a single weakness, while the designer must find and eliminate all weaknesses to achieve perfect security.
7. There is a natural tendency on the part of users and system managers to perceive little benefit from security investment until a security failure occurs.

Challenges in Computer Network and Information security (Cont....)

8. Security requires regular, even constant, monitoring, and this is difficult in today's short-term, overloaded environment.
9. Security is still too often an afterthought to be incorporated into a system after the design is complete rather than being an integral part of the design process.
10. Many users and even security administrators view strong security as an impediment to efficient and user friendly operation of an information system or use of information. The difficulties just enumerated will be encountered in numerous ways as we examine the various security threats and mechanisms.
11. Internet infrastructures are vulnerable.
12. Solutions usually require a larger scale of modification
13. Security and performance tradeoff
14. Security is only as strong as the weakest link
15. Attacks can be easily launched and difficult to be traced

What are the basic security measure?

1. External security: is about protection of computer systems from environmental damages such as floods and heat, physical security such as locking rooms and computers and electrical protection such as power surge and electromagnetic interfaces.
2. Operational security: is about deciding who has access to what, limiting access time and location
3. Surveillance: proper placement of a security cameras and can deter theft and vandalism
4. Passwords/authentication: the most common form of security. There are some simple rules for password security systems like:
 1. Change your password often
 2. Pick a good, strong and random password
 3. Don't share password or write down them
 4. Don't select/use names and familiar objects as password

What are the basic security measure? (Cont....)

- ✈ Authentication: the process of reliably verifying the identity of someone/something by means of a secret (password), an object (smart card), physical characteristics (finger print) and trust.
- 5. Auditing: used to detect wrong doing
- 6. Access rights: determines the security by means of who and how. Who do you give access right to? (No One, Group of Users, entire set of Users). How a user/group of user does has access? (Read, write, delete, print, copy, execute).
- 7. Standard System attacks
- 8. Viruses/worms and antivirus tools
- 9. Firewalls
- 10. Encryption and Decryption Techniques
- 11. Digital Signature
- 12. Security Policy

History of Computer Network and Information Security

1. Discuss about the history of computer networks and Information security in chronological order. Take one computer network and information security Model and discuss about it. (The best feature of the Model, the basic vulnerabilities about the model and the application area of the security model.)

Security/Privacy Vulnerability

✚ **Vulnerabilities(Attack Surface):-** are weak points or loopholes in security that an attacker can exploit in order to gain access to the network or to resources on the network.

✚ The vulnerability is not the attack, but rather the weak point that is exploited.

✚ Vulnerability is the intersection of three elements:

1. A system susceptibility or flaw,
2. Attacker access to the flaw, and
3. Attacker capability to exploit the flaw

- To be vulnerable, an attacker must have at least one applicable **tool or technique** that can connect to a system weakness.
- A security risk may be classified as a vulnerability. But there are **vulnerabilities without risk**, for example when the affected asset has no value.

✚ A vulnerability with one or more known instances of working and fully-implemented attacks is classified as an exploitable vulnerability, a vulnerability for which an exploit exists.

Vulnerability Classification

✉ Vulnerabilities are classified according to the asset class they related to:

1. Hardware

- susceptibility to humidity
- susceptibility to dust
- susceptibility to soiling
- susceptibility to unprotected storage

2. Software

- insufficient testing
- lack of audit trail

3. Network

- Unprotected communication lines
- Insecure network architecture

4. Personnel

- Inadequate recruiting process
- Inadequate security awareness

5. Site

- Area subject to flood
- Unreliable power source

6. Organizational

- Lack of regular audits
- Lack of continuity plans

Tips for computer and network security

- ✧ Use a good anti-virus program.
- ✧ Make sure your virus definitions are up-to-date.
- ✧ Run regular virus scans.
- ✧ Update your operating system regularly.
- ✧ Configure and use a firewall.
- ✧ Use your Web browser's security features.
- ✧ Enable your router's security features.
- ✧ Install an anti-spyware program.
- ✧ Use strong, varied passwords.
- ✧ Consider a computer network security suite and policy

Assignments and Exercise

- ❖ Read more about on these topics from the text books and other legitimate sources.
- ❖ Your individual and Group Assignment will be posted on E-learning as soon as the chapter content is completed.
- ❖ Cryptography and Network Security: Principles and Practice, Sixth Edition, or Latest editions